Visual Diagnostics for NEB Paths

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> *Smooth barriers don't certify path convergence.*

Background

Most NEB mistakes I care about start after the run looks plausible. The energy profile smooths out, the climbing image sits near a barrier, and the directory has enough `neb_*.dat` files to look settled. That stage needs plots that keep image motion, residual forces, and geometric detours visible.

These notes sit next to the transition-state work on my research site. The method names follow the published OCI-NEB paper (Goswami, Gunde, and Jónsson 2026). The two-dimensional panels follow the RMSD projection paper (Goswami 2026b). The workflow follows the NEB orchestration paper and the Atomistic Cookbook eOn + PET-MAD recipe (Goswami 2026a).

Most figures use the 1,3-dipolar cycloaddition of ethylene and nitrous oxide to form 4,5-dihydro-1,2,3-oxadiazole. The system fits on a web page while still running through PET-MAD (Mazitov et al. 2025) through metatomic (Bigi et al. 2026), eOn (Chill et al. 2014), energy-weighted CI, and OCI-NEB. I add Grignard and bicyclobutane later because those paths turn. Windows packaging snags that show up next to these scientific builds are collected in Windows Compatibility and Scientific Computing.

Figure reproduction

The companion repository carries the pinned environment and the Snakemake workflow. The post shouldn't duplicate an environment file that drifts away from the run. The repository carries that contract.

```
git clone https://github.com/HaoZeke/nebviz_repro
cd nebviz_repro
pixi run -e cpu dry-run
pixi run -e cpu all
```

The `pixi` tasks call Snakemake under `eonRuns`:

```
cd eonRuns
snakemake --configfile config/general_config.yml -n -p
snakemake --configfile config/general_config.yml -c4 all_figures
```

The `all_figures` target exports PET-MAD, prepares and minimizes endpoints, writes eOn input files, runs OCI-NEB for three systems, validates the resulting `.con` files with readcon v2, and writes figures to `eonRuns/results/03_figures/blog`. The eOn configuration includes `ci_mmf = true` and the dimer block used by OCI-NEB. `stack.json` records `eonclient` 2.14.0 at commit 4a52123, readcon 0.8.0, rgpycrumbs 1.7.0, chemparseplot 1.7.0, NumPy 2.4.2, and matplotlib 3.10.9.

The plotted inputs stay small enough to audit: the final band (`neb.con`), per-step energy/force data (`neb_*.dat`), and path snapshots (`neb_path*.con`). Profile and landscape figures come from `python -m rgpycrumbs.cli eon plt-neb`. The three table diagnostics read the eOn tables directly.

This repository also carries a figure reproduction bundle under `repro/neb-visualizations/`. It is not the full computational benchmark. The committed CSV files are reduced exports from the NEB outputs. They contain the data needed for the blog figures. Run `pixi run -C repro/neb-visualizations figures` to regenerate the PNGs from the tables.

One dimension, three coordinates

The usual NEB profile already encodes a decision: what counts as horizontal distance. I draw three scalar views. Path length gives the conventional barrier. Image index makes bands with different image counts easier to compare. RMSD from the reactant puts the horizontal coordinate back in Angstroms after IRA alignment.

```
python -m rgpycrumbs.cli eon plt-neb \
  --con-file results/02_neb/system_100/neb.con \
  --output-file results/03_figures/blog/neb-viz-1d-path.png \
  --plot-type profile --rc-mode path --plot-structures crit_points \
  --strip-renderer xyzrender --perspective-tilt 8 --rotation 15x,55y,0z
```

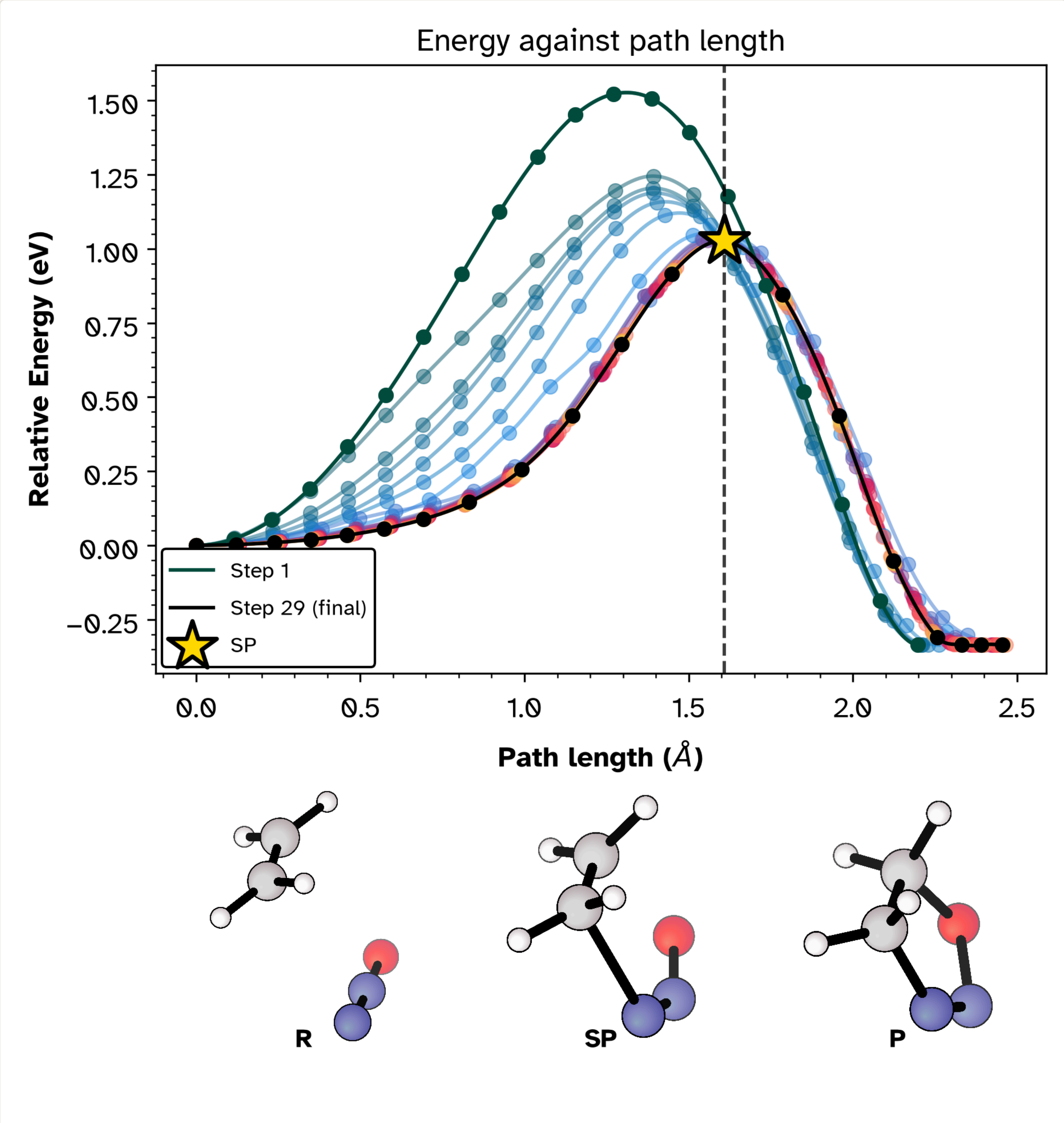

Figure 1: Energy against path length. The peak gives the barrier. The piecewise cubic Hermite spline uses the parallel forces, so the curve respects dE/ds at every image instead of wiggling between points. Dark to light traces the optimization. The converged path is black; R, SP, and P sit in a strip under the panel rather than on the curve.

For the cycloaddition, the path-length view tells the expected first story: the barrier falls from about 1.52 eV on the initial band to about 1.03 eV by step 29, and the product sits about 0.34 eV below the reactant. It can't show sideways motion, sparse sampling, or whether the high image sits in a narrow channel.

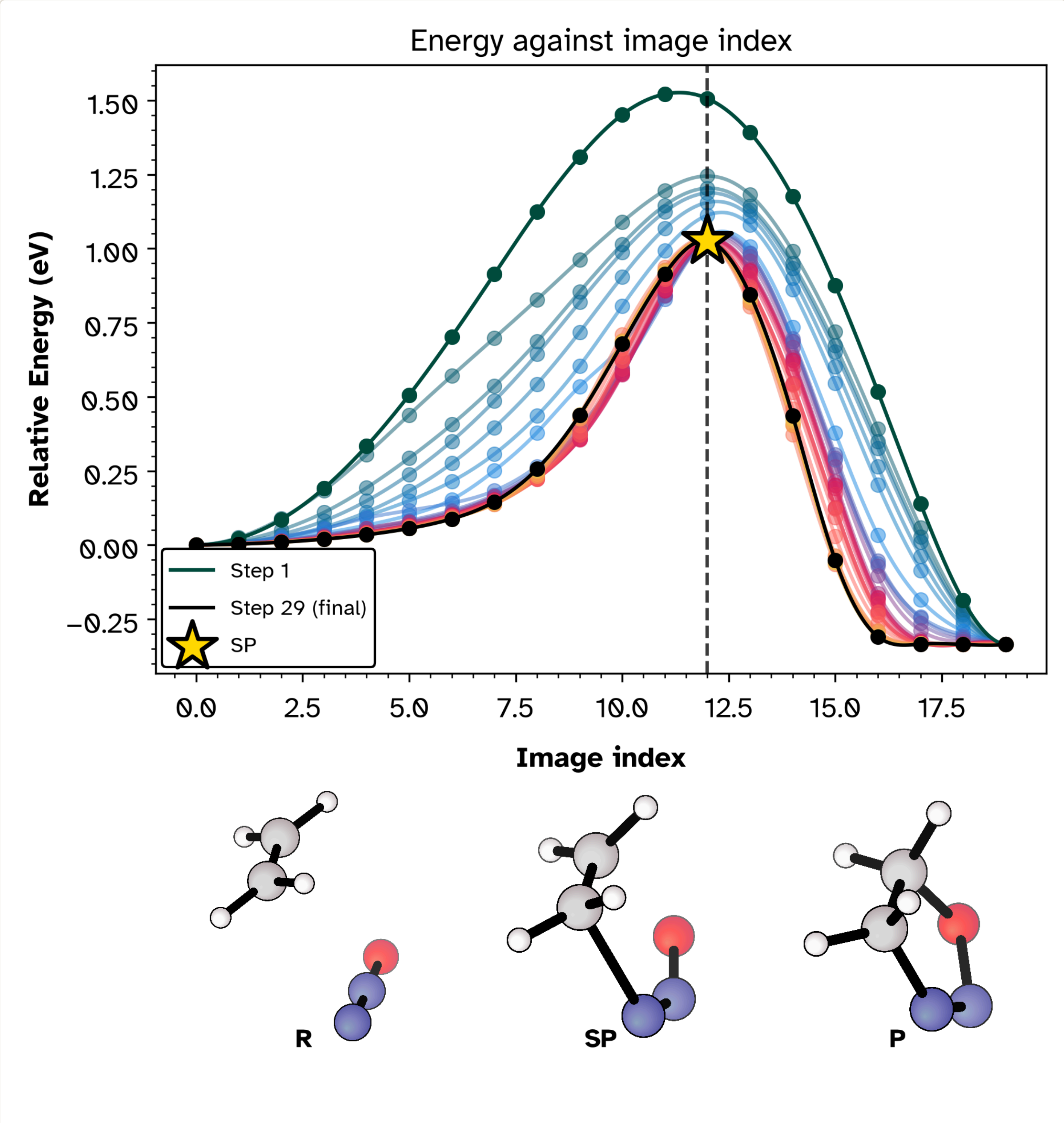

Figure 2: Energy vs image index. Even spacing makes bands of different image counts comparable. Smooth cubic through the energies (not Hermite: force slopes are not dE/d(index)). Same strip layout as the path-length panel.

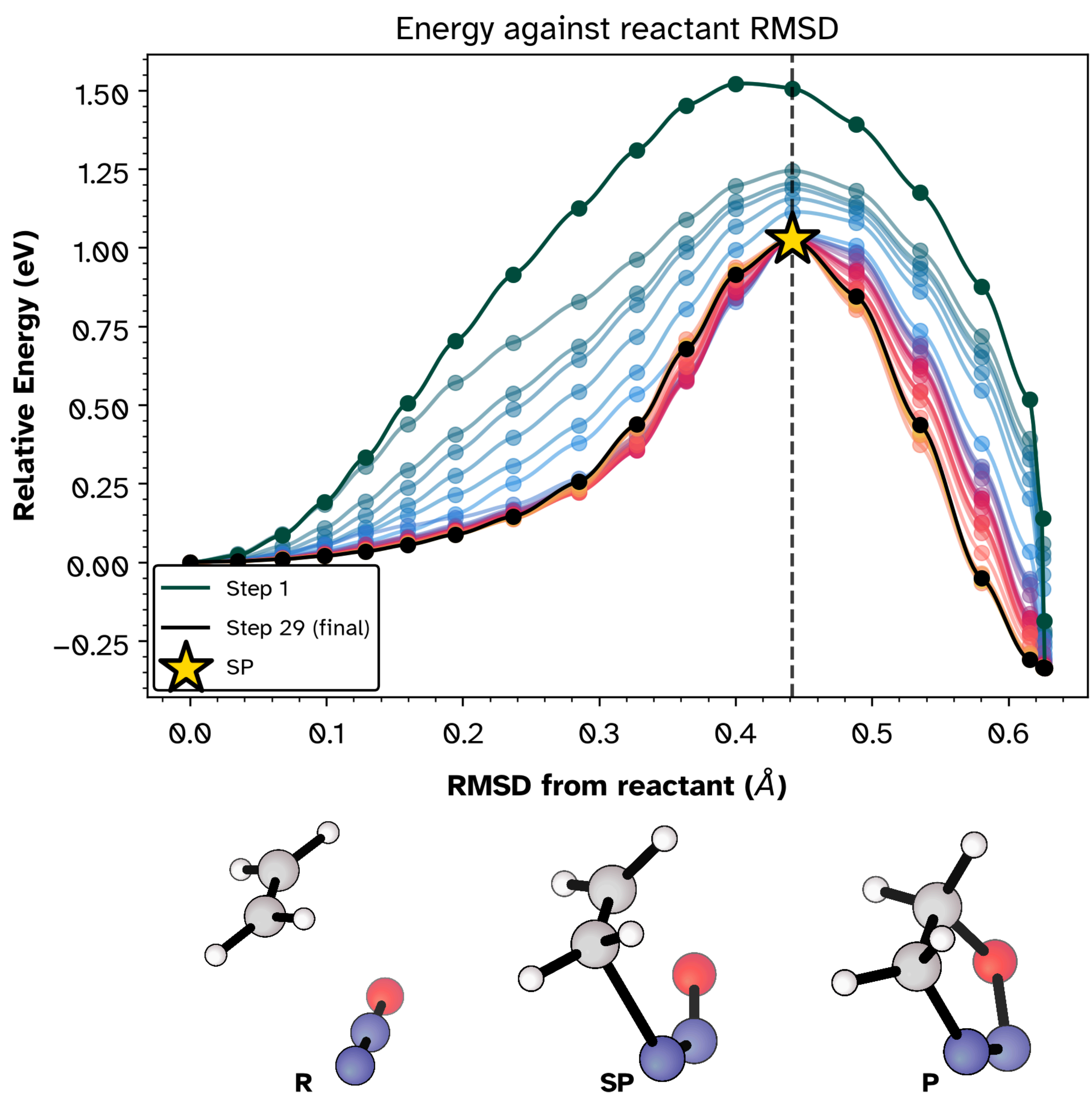

Figure 3: Energy vs RMSD from the reactant: the coordinate becomes Angstroms of real geometric change. Structures remain below the axes so the curve stays readable.

I keep all three because they fail differently. Path length can look settled while image index shows migration. RMSD can expose geometric motion that path length folds into arc length.

Projection into two dimensions

The 2D projection makes the geometric check explicit. I measure each structure by IRA RMSD to the reactant and product, giving coordinates r and p. The NEB parallel forces project into the same plane as synthetic gradients. Energies and gradients train a gradient-enhanced Gaussian process with an inverse-multiquadric kernel. A final rotation maps r, p onto reaction progress s and orthogonal deviation d. The RMSD projection paper gives the derivation and failure modes (Goswami 2026b).

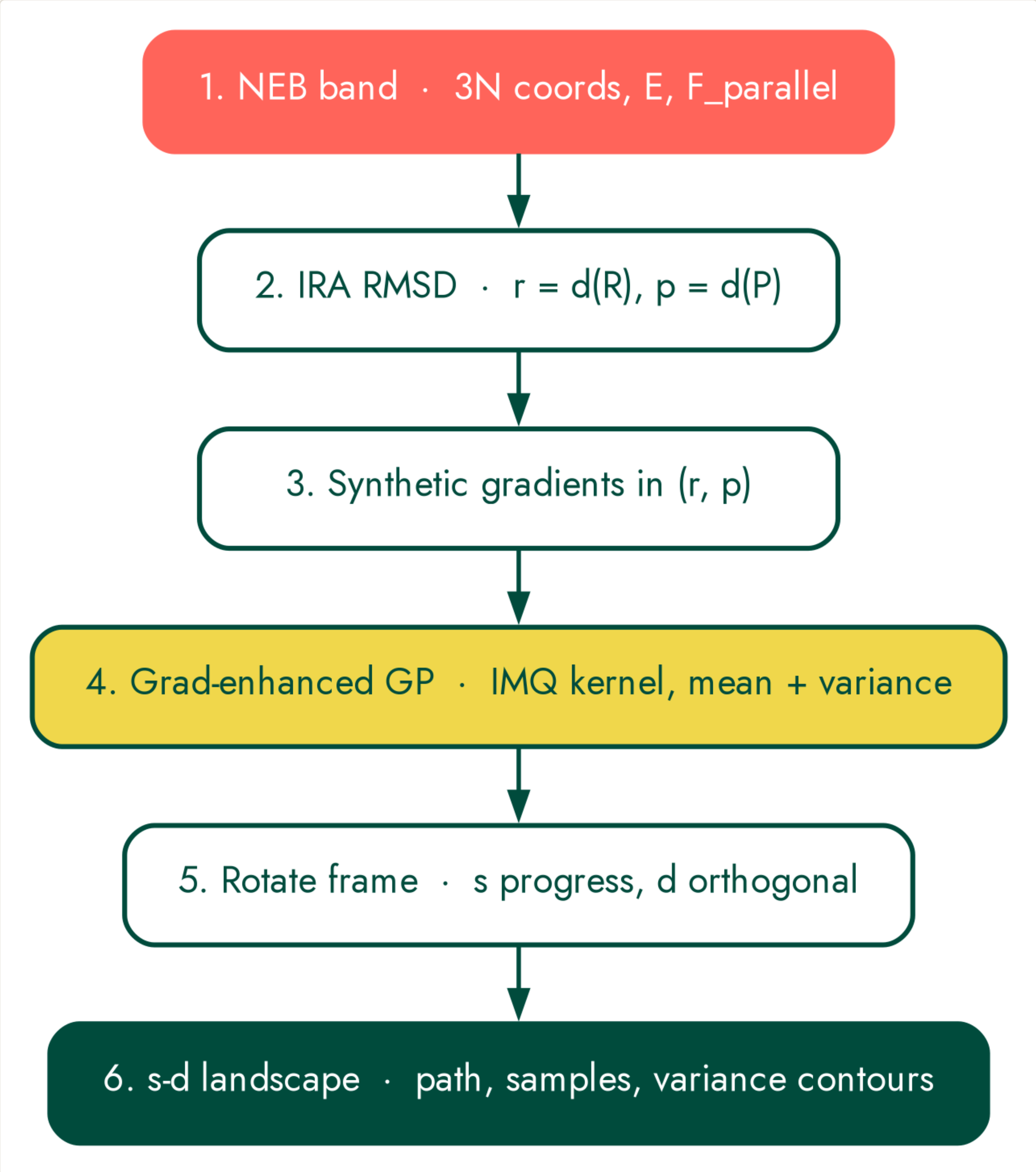

Figure 4: Projection pipeline from 3N-dimensional NEB output to IRA RMSD coordinates, synthetic gradients, a gradient-enhanced GP surface, and the rotated s-d frame.

```
python -m rgpycrumbs.cli eon plt-neb \
  --con-file results/02_neb/system_100/neb.con \
  --output-file results/03_figures/blog/neb-viz-2d-cyclo.png \
  --plot-type landscape --project-path --show-pts --landscape-path all \
  --plot-structures crit_points --surface-type grad_imq --ira-kmax 14 \
  --title Cycloaddition --theme ruhi --cmap-landscape ruhi_diverging \
  --strip-renderer xyzrender --perspective-tilt 8 --rotation 15x,55y,0z \
  --show-legend --additional-con resources/ewNEB/configurations/system100-sp.xyz \
  ORCA
```

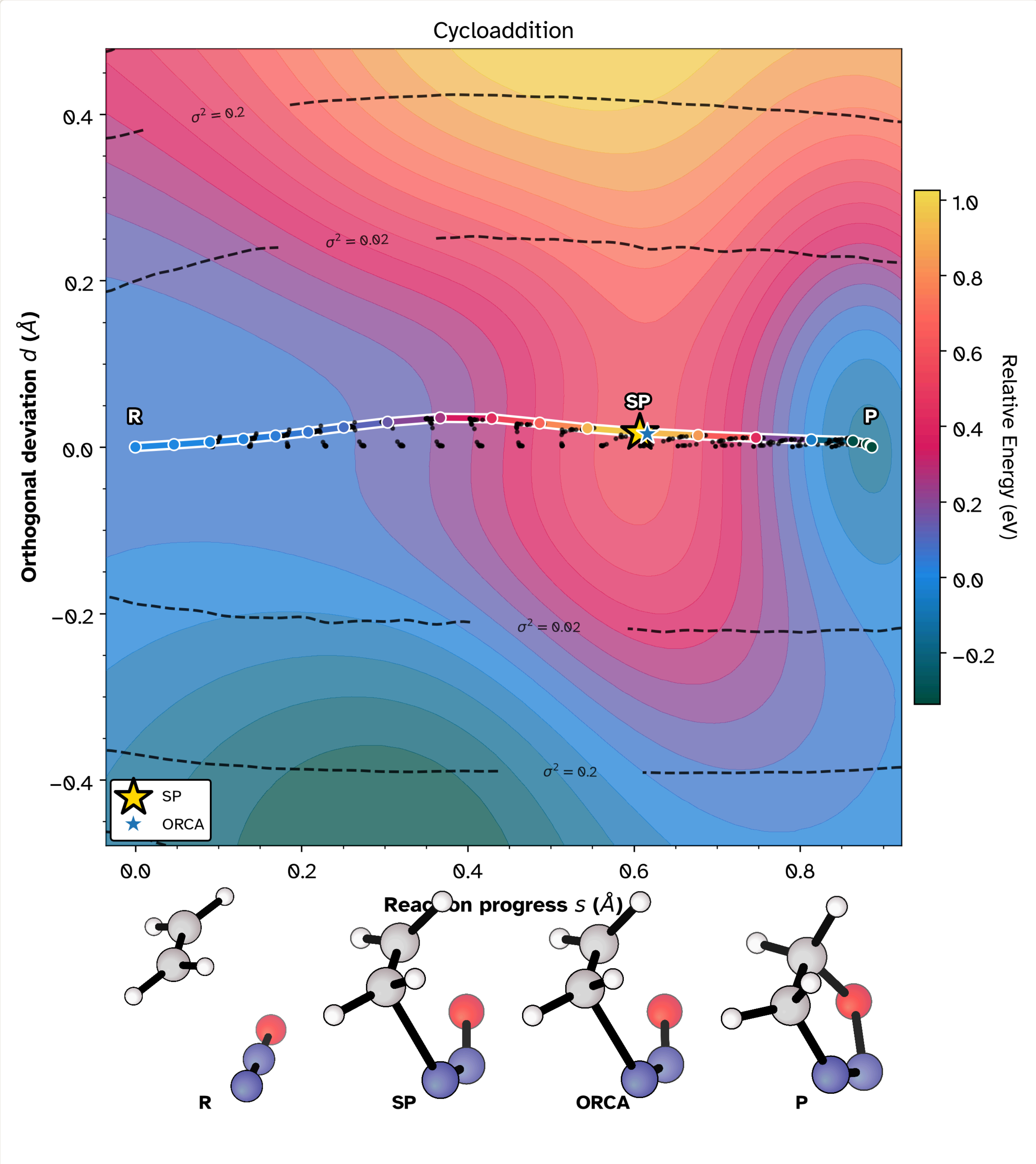

Figure 5: Cycloaddition in the s-d frame. Ruhi energy surface, sampled structures as black dots, converged path as open circles, posterior-variance contours as dashed curves, and the ORCA B3LYP-D3 saddle as a star (strip label: ORCA).

The projection does more than recolor the profile. The color field, sampled points, force-derived gradients, posterior variance, path history, and reference saddle share one coordinate system. Here the black points hug the converged path, so the optimization sampled one channel rather than a broad cloud. The independent ORCA B3LYP-D3 saddle (Ásgeirsson et al. 2021) sits in the same barrier region as the PET-MAD/eOn path. The dashed variance contours shrink near sampled structures and widen at the edge, marking data support for the interpolation. A scalar barrier can say that two calculations differ. The projection says where.

When the path turns

The cycloaddition stays close to the diagonal. The Grignard addition (Birkholz and Schlegel 2015) and the bicyclobutane ring opening from the mlFSM work (Marks and Gomes 2025) give better stress tests. In the s, d plane both paths bow away from the reactant-product diagonal. The 1D coordinate pays for that turn by stretching arc length and hiding direction.

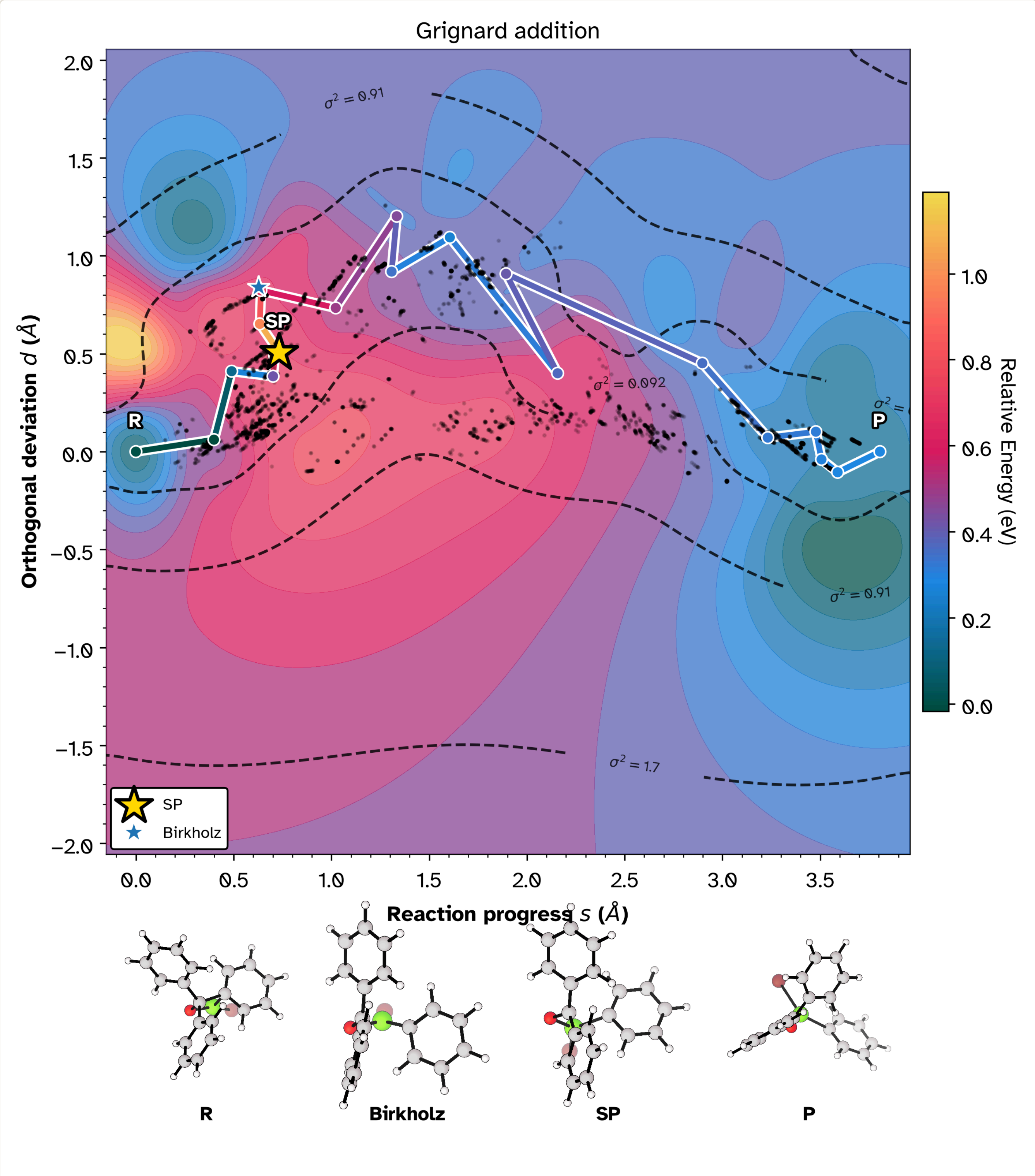

Figure 6: Grignard addition. The path bows off the reactant-to-product line in the s-d plane; Birkholz's reference saddle sits near the band SP. The 1D coordinate folds that turn into arc length.

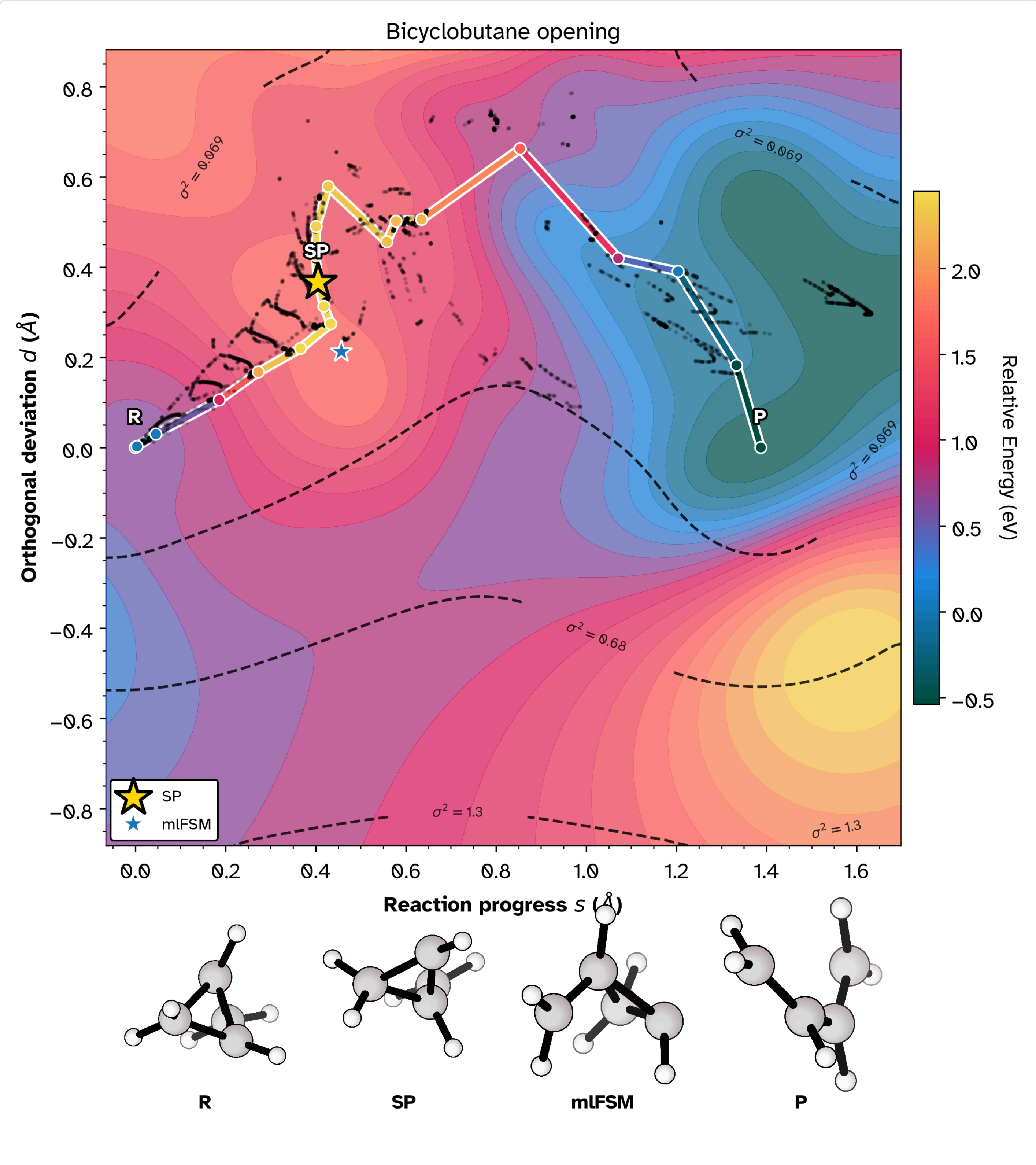

Figure 7: Bicyclobutane ring opening. A geometric kink shows up as off-diagonal motion; the mlFSM reference saddle is marked separately from the band SP.

Table diagnostics

Surfaces carry more signal when the force tables remain nearby. The Grignard case has force spikes, MMF back-off steps, and dimer mode loss before it settles enough for the visualization target. That normal reaction-path ugliness belongs in the reproducible record. The table plots answer narrower questions: did the barrier stabilize, which images still move, and how evenly did the final images spread out?

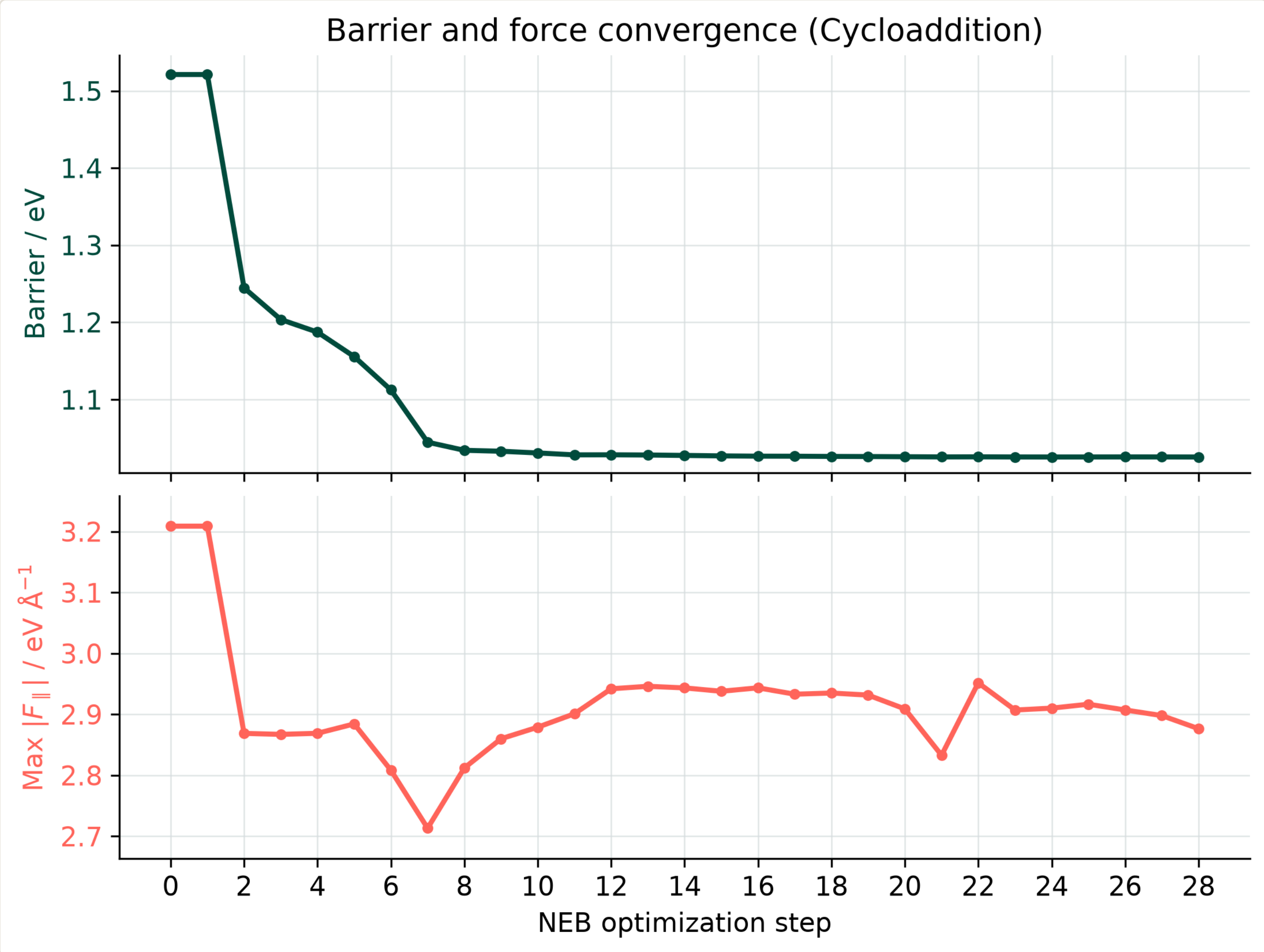

Figure 8: Cycloaddition barrier (top) and max parallel force (bottom) over all 29 NEB steps. The barrier settles near 1.03 eV while the force stays elevated: a flat barrier is not force convergence.

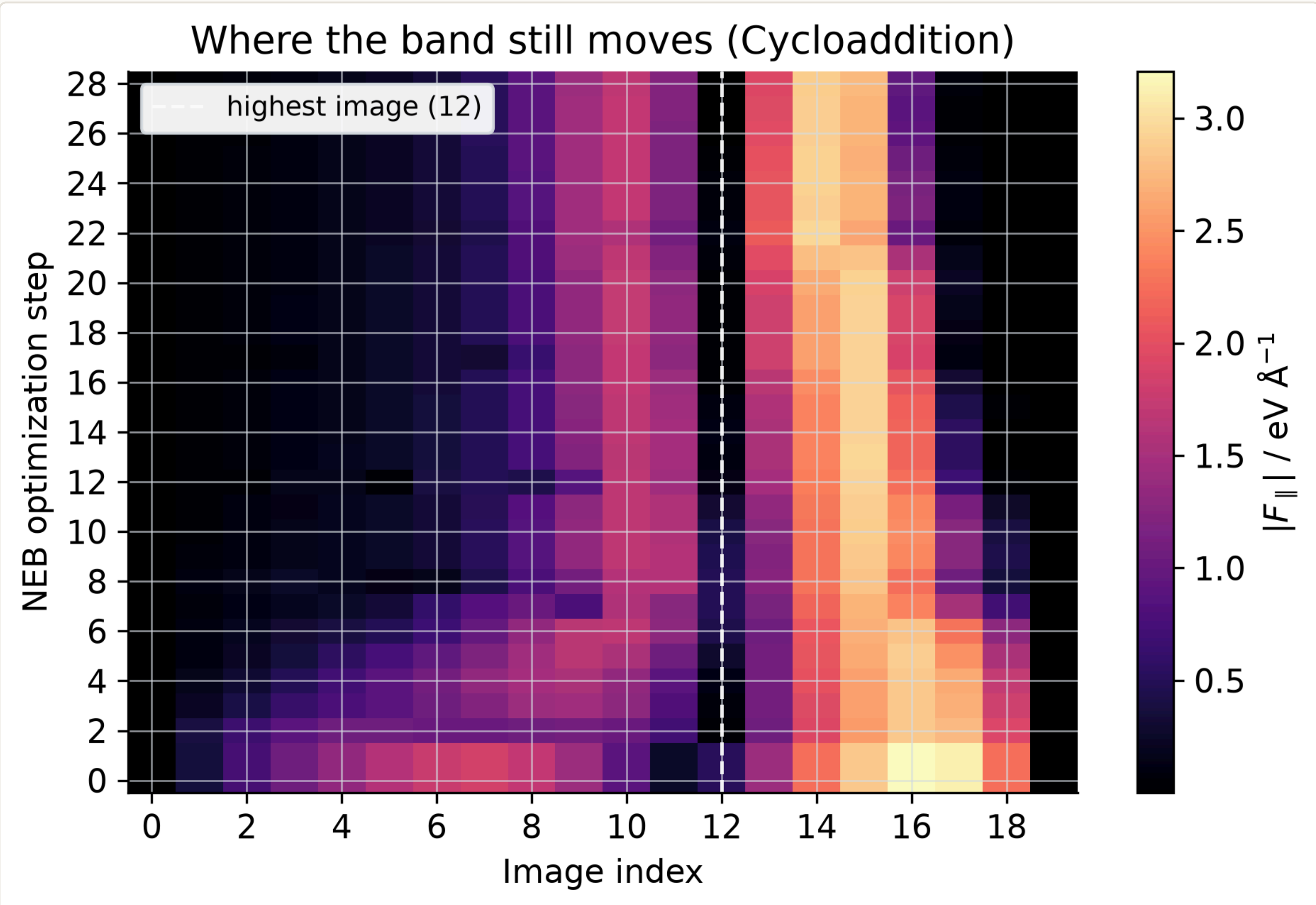

Figure 9: Parallel-force heatmap by image and optimization step for the cycloaddition. Integer image indices; dashed line marks the highest-energy image. Late motion sits product-side of the climb, not only at the top image.

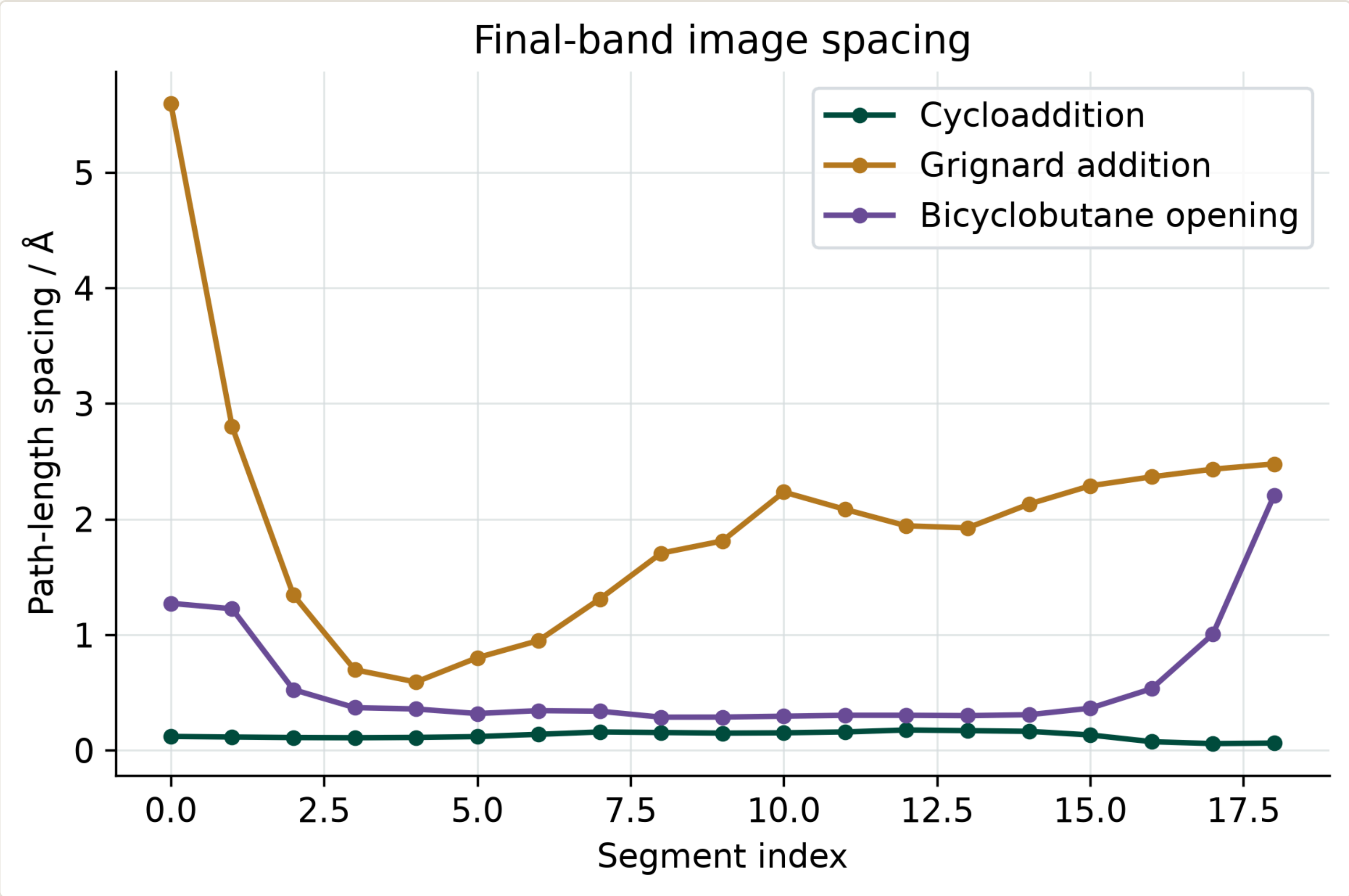

Figure 10: Final-band path-length spacing for the three example systems. Uneven spacing need not mean a wrong path, but it changes what a 1D profile can claim.

Method labels

The figures say OCI-NEB because that name appears in the published enhanced CI-NEB method with Hessian eigenmode alignment (Goswami, Gunde, and Jónsson 2026). Don't read it as a plotting label. Plain CI-NEB drives the highest image uphill along the band tangent (Henkelman, Uberuaga, and Jónsson 2000). It can locate the right region and still spend the slow part near the top, or stall near a flat mode rather than a proper first-order saddle.

OCI-NEB keeps the band context and switches the top image into MMF when the curvature information says the transition basin deserves a saddle search. In the Frontiers paper, Baker-Chan/PET-MAD runs used 57 percent fewer energy and force calls at the median, with a 95 percent credible interval from 50 to 64 percent. The Pt 111 heptamer set used 31 percent fewer calls across 59 transitions. Fixed switching at 0.5 eV/A spent 46 percent more calls than the adaptive criterion. Those numbers don't make NEBs forgiving. They say the expensive part often lies in deciding when the top image can take over saddle work.

Related publications

- The s, d frame: (Goswami 2026b)
- OCI-NEB and the call-count numbers: (Goswami, Gunde, and Jónsson 2026)
- The Snakemake / eOn lifecycle: (Goswami 2026a)

The PNGs on this page come from the reduced tables in `repro/neb-visualizations/`.

Conclusions

The check I want after an NEB run has four parts: scalar profiles for the barrier, RMSD coordinates for geometry, a two-dimensional projection for path support, and force/spacing tables for optimizer behavior. The cookbook gets the eOn/PET-MAD calculation onto disk. The companion repository regenerates the figures and records the stack. The papers define the methods.

Reading list

Papers behind the diagnostics here, the OCI-NEB and RMSD-projection methods, the potentials, and the codes that run them, collected as a citation list: 11 references. The collection exports to BibTeX or any CSL style, so it can go straight into a bibliography.

References

Ásgeirsson, Vilhjálmur, Benedikt Orri Birgisson, Ragnar Bjornsson, Ute Becker, Frank Neese, Christoph Riplinger, and Hannes Jónsson. 2021. "Nudged Elastic Band Method for Molecular Reactions Using Energy-Weighted Springs Combined with Eigenvector Following." *Journal of Chemical Theory and Computation* 17 (8): 4929--45. <https://doi.org/10.1021/acs.jctc.1c00462>.

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---------. 2026b. "Two-Dimensional RMSD Projections for Reaction Path Visualization and Validation." *MethodsX* 16 (March): 103851. <https://doi.org/10.1016/j.mex.2026.103851>.

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Marks, Jonah, and Joseph Gomes. 2025. "Incorporation of Internal Coordinates Interpolation into the Freezing String Method." *Journal of Chemical Theory and Computation* 21 (23). <https://doi.org/10.1021/acs.jctc.5c01492>.

Mazitov, Arslan, Filippo Bigi, Matthias Kellner, Paolo Pegolo, Davide Tisi, Guillaume Fraux, Sergey Pozdnyakov, Philip Loche, and Michele Ceriotti. 2025. "PET-MAD as a Lightweight Universal Interatomic Potential for Advanced Materials Modeling." *Nature Communications* 16 (1): 10653. <https://doi.org/10.1038/s41467-025-65662-7>.